

Experiment 9: GCD (Greatest Common Divisor)

Aim

To implement the **Euclidean Algorithm** in Python to find the **Greatest Common Divisor (GCD)** of two numbers.

Algorithm

Step 1: Start.

Step 2: Read two integers a and b.

Step 3: While b is not equal to 0:

- Find the remainder of a divided by b.
- Assign a = b.
- Assign b = remainder.

Step 4: When b becomes 0, a is the GCD.

Step 5: Display the GCD.

Step 6: Stop.

Source Code (Python)

main.py

```
1 # GCD using Euclidean Algorithm
2
3 a = int(input("Enter first number: "))
4 b = int(input("Enter second number: "))
5
6 while b != 0:
7     a, b = b, a % b
8
9 print("GCD =", a)
```

Output

Output

```
Enter first number: 48
Enter second number: 18
GCD = 6
```

```
=== Code Execution Successful ===
```

Time Complexity

- **Best Case: $O(1)$**
- **Average Case: $O(\log n)$**
- **Worst Case: $O(\log n)$**

Space Complexity

- **$O(1)$**

Result

The Greatest Common Divisor (GCD) of two numbers was successfully found using the Euclidean Algorithm in Python. The program correctly computes and displays the GCD of the given numbers.